SOL PROTOCOL

Playable Tutorial

Flow of the on-foot and ship tutorials, and setup of every tutorial step

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1 Overview

The playable tutorial teaches the core mechanics of Sol Protocol in two parts. The **on-foot tutorial** takes place inside a training dungeon and covers movement, comfort settings, gravity floors, zero-gravity movement, the personal inventory and grabbing objects. The **ship tutorial** takes place aboard the Pathfinder and covers the ship inventory, energy management, the gunner station, shields, piloting, repairs and the warp jump.

Each step shows a tutorial panel (title and text) and a guide line that points to the next target. The step ends when its completion condition is met, and the next step starts.

1.1 At a glance

Part	Where	Steps	What the player learns
On foot	Training dungeon	F1–F22	Movement and camera, comfort options, buttons, crouch, jump, gravity floors, zero-gravity flight, personal inventory, grabbing and storing loot, airlock
Ship	Pathfinder, cockpit and an asteroid field	S1–S34	Ship inventory, energy management, gunner station and turret, automa, shields, holomap, gearbox, flight controls, indicators, repairs, warp jump

1.2 How to read the step tables

Column	Meaning
#	Step ID: **F** = on foot, **S** = ship. The tutorial flow below uses the same IDs.
Step	Panel title as shown in game (wrapped in `<h2>…</>` in the game text) and the widget blueprint. Blueprint names omit the `WB_Tutorial_` prefix.
Player text	Panel text. Words in **bold** are wrapped in `<Key>…</>` in the game text.
Completes when	What the player must do to finish the step and start the next one.
Trigger / pointer	Trigger actor, target point or pointer tag. `PlaneXX` and `SphereXX` stand for `BP_TutorialTrigger_PlaneXX` and `BP_TutorialTrigger_SphereXX`.

Column	Meaning
Task	State machine task, without the `STTask_Tutorial_` prefix.
Line target	What the guide line points at.

2 Tutorial flow

2.1 On-foot tutorial

The map below shows the route through the dungeon. The numbers on the map match the numbered blocks that follow.

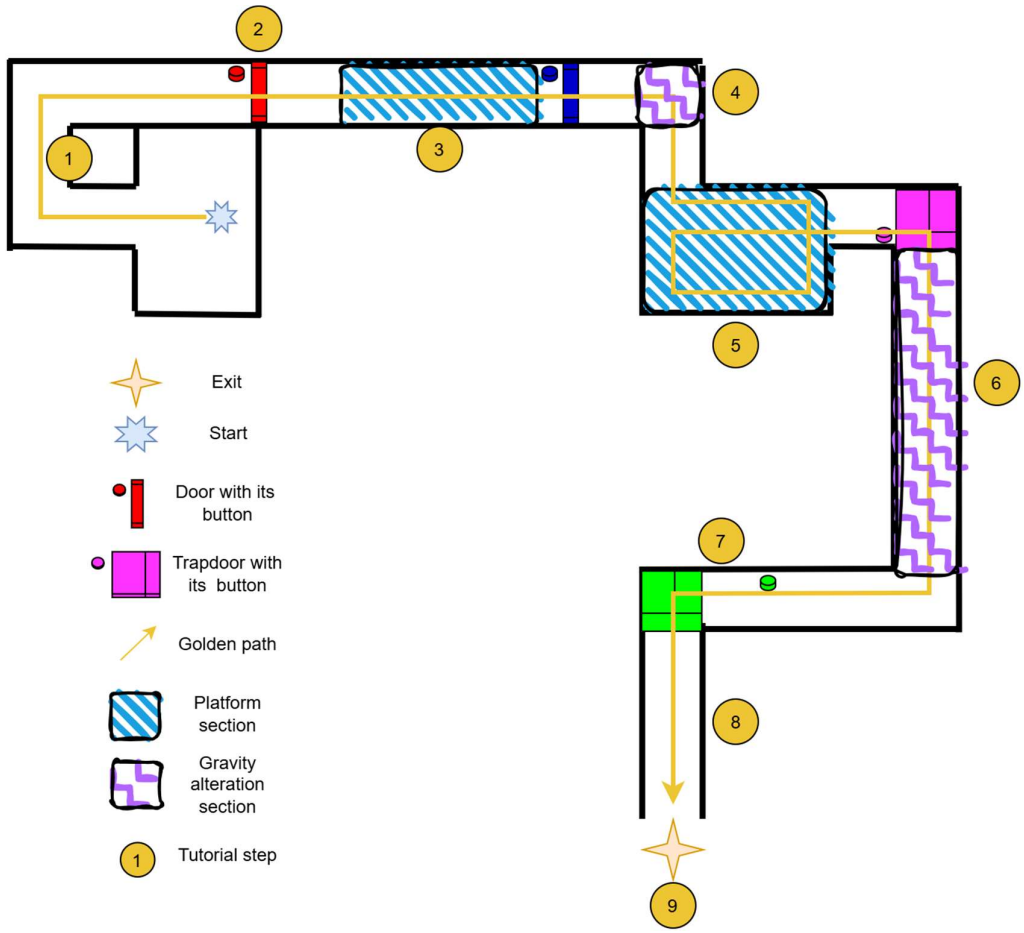


Figure 1 *On-foot tutorial map: golden path from the start room to the airlock, with doors, trapdoors, platform and gravity sections.*

1 Basics F1–F2

The player starts on foot inside a sealed room. They learn to move with the left stick and turn the camera with the right stick. The tutorial also points to the comfort options: snap or smooth turning, and the vignette.

2 First door F3

The room contains a door and a button. The player pushes the button with the index finger to open the door.

3 Crouch and jump F4–F6

Beyond the door, a pipe blocks the corridor: the player crouches (B) to pass under it. If their height feels wrong, they can recalibrate the seated height offset from the options.

The corridor continues with four jumps (A) that lead to the second door. The player already knows how to open doors, so there is no panel for it: the next step starts when they cross the threshold.

4 Gravity floors F7–F8

A vertical corridor with gravity floors carries the player two levels up, walking on the walls. The tutorial warns that turning off the fade on gravity changes can increase motion sickness, and shows that to leave a gravity floor the player simply walks off it.

5 Combined section F9

An L-shaped corridor, two units high, with five jumps, a gravity floor and a pipe, combines all the movement learned so far.

6 Zero gravity F10–F15

In a long, multi-level room, a button right in front of the player turns gravity off. The player learns the zero-gravity controls: take off (A), move (left stick), rise, descend and turn (right stick), brake (B).

A button opens the first trapdoor, leading to the zero-gravity parkour zone above. There, the player learns to roll (press a stick) and change pitch (X/Y), and uses all these moves to get past the obstacles.

7 Gravity back on F16–F17

At the end of the course, a button turns gravity back on and opens a second trapdoor. The player drops through it into the rest of the dungeon; the fall does no damage.

8 Loot room F18–F20

The player enters a loot room and learns the personal inventory: it holds up to 4 items and opens from the left wrist with the right hand. They learn to grab objects (one per hand, without losing mobility) and store the missile launcher ammo box by releasing it over the inventory button. The inventory does not need to be open to store items.

9 Back to the ship F21–F22

The player opens the dungeon airlock, steps out into space and uses the zero-gravity controls to fly back to the ship and board it.

2.2 Ship tutorial

The map below shows where each part of the ship tutorial takes place. The numbers on the map match the numbered blocks that follow: blocks 1–3, 5 and 6 all happen at the start point.

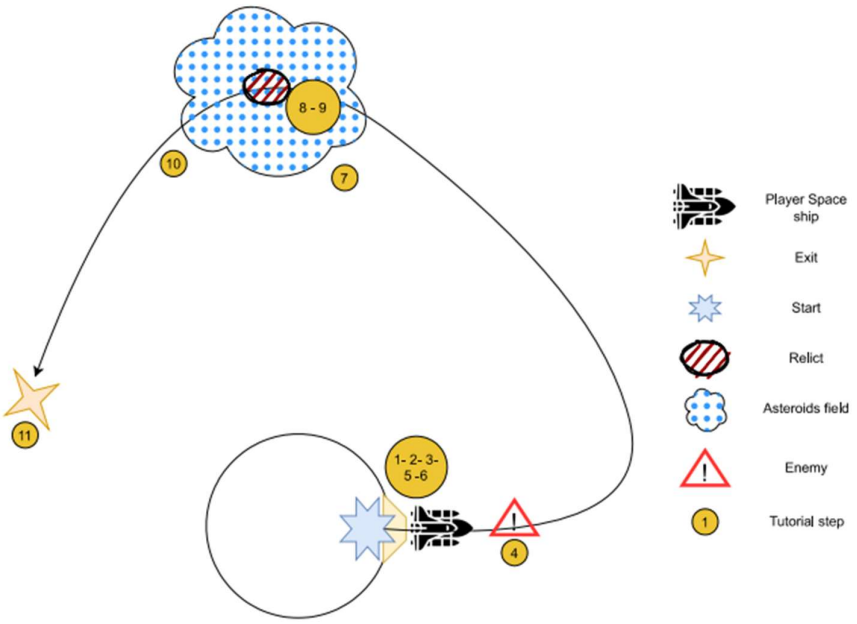


Figure 2 *Ship tutorial map: start, enemy encounter, asteroid field with the relic, and the warp exit.*

1 Storing the loot S1–S3

Aboard the Pathfinder, the player takes the missile launcher ammo box out of the personal inventory (grabbing it extracts it straight into the hand), places it on the ship inventory table and presses Insert. Stored ammo is used automatically by the weapons that need it.

2 Captain's station S4–S7

Mines are blocking the way. The player walks through the cockpit door, sits in the captain seat and learns energy management: clicking the squares next to a system assigns energy units (1 yellow square = 1 unit). They assign 5 units to the weapons, then stand up with the seat belt button or A.

3 Gunner's station S8–S11

The player sits in the gunner seat and puts on the Sol headset to enter turret view. They assign energy to the gatling and/or missile launcher, aim with the head, rotate the turret with the right stick, fire the front turrets with the triggers and the side guns with the sticks. They also learn the health bar and hit feedback on target indicators. The step ends when all mines are destroyed.

4 Enemy incoming S12–S14

A large enemy ship warps in. The player takes off the headset, stands up and activates the automa with the button behind the gunner seat. While active, the automa fires the guns automatically at enemies in range.

5 Shields S15–S19

From the ship inventory panel, the player extracts the shield system chip and inserts it into the matching system panel. Back in the captain seat, they move 3 energy units from the weapons to the shields (3 on shields, 2 on weapons). Using the handle on the holomodel, they move the shield around the ship to block hits, shown by red triangles, while the automa destroys the enemy ship.

6 Piloting S20–S24

With the enemies gone, the player moves all 5 energy units to the engine and goes to the pilot station. They learn the holomap (zoom with the cursor, ping objects, set a course to the marked asteroid field), the gearbox (inertial brake, operational flight, micro-thrust) and the flight controls (triggers to accelerate and decelerate, sticks to rotate). The holomodel shows rotation and speed, and more energy in the engines means more speed.

7 Asteroid field S25–S27

The player flies to the asteroid field and learns the indicators: red triangles are enemies, red rhomboids are enemy ship parts (weak points), green hexagons are loot. Following the loot indicator to a structure on an asteroid, they leave the ship and grab a hull repair sphere.

8 Repair S28–S29

Back aboard, the player places the sphere in the round hole of the system panel, which absorbs it and repairs the hull. The tutorial also mentions the blue system repair sphere, which repairs systems. With the hull repaired, the ship has more energy available for the warp.

9 Warp preparation S30–S31

In the captain seat, the player assigns 5 energy units to the warp, which only works when fully powered, and leaves at least 1 unit on the engine. At the galactic map, they choose a sector and select Calculate coordinates.

10 Warp jump S32–S33

Back at the pilot station, the player flies out of the asteroid field to clear the path: anything hitting the ship during the warp may destroy it. They align the ship with the luminous trail until it turns green, hold position and push the holo lever forward to jump.

11 End S34

After the jump, the tutorial is complete and the player returns to the lobby.

3 Steps: on foot

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
F1	**MOVEMENT** `Movement`	Use the **left stick** to move around.	Collider	`Plane00` · tag `Movement`	`TriggerPlane`	—
F2	**MOVE THE CAMERA** `Camera`	Use the **right stick** to rotate the camera. If this is uncomfortable, press **Select** to open the menu. In **Options**, choose **snap** or **smooth turning**. In **Comfort**, enable the **vignette** and adjust its intensity.	Collider	`Plane01` · tag `Camera`	`TriggerPlane`	—
F3	**INTERACT WITH BUTTONS** `Interact`	Use your **index finger** to push the **button** down.	Button pressed	`Door01` TargetPoint `Door1Button`	`StateValue`	First button (`Door1Button`)
F4	**CROUCH** `Crouch`	Press **B** to **crouch**. If your height feels wrong, press **Select** and open **Options**. Adjust the **seated height offset**, then press **Calibrate**.	Collider	`Plane02` · tag `crouch`	`TriggerPlane`	—
F5	**JUMP** `Jump`	Press **A** to **jump**.	Collider	`Plane02_5` · tag `jump`	`TriggerPlane`	—
F6	**REACH THE NEXT ROOM** `WP_Tutorial_Transition`	Proceed to the next section.	Player crosses the second doorway (collider) *Note: the door is there, but the player already knows how to open doors, so the next step starts at the threshold.*	`Plane03`	`TriggerPlane`	—

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
F7	**GRAVITY FLOORS** `GFloor`	**Gravity floors** change the gravity for your character. Step onto one to walk on walls normally. For a more immersive experience, go to **Options › Comfort** and disable **Fade when gravity changes**. **Disabling this setting can significantly increase motion sickness.**	Collider	`Plane03_5`	`TriggerPlane`	—
F8	**LEAVE GRAVITY FLOORS** `GFoorExit`	To leave a gravity floor, simply walk off it.	Collider	`Plane04`	`TriggerPlane`	—
F9	**REACH THE NEXT SECTION** `Parkour`	Proceed to the next room.	Collider	`Plane05` · tag `Parkour`	`TriggerPlane`	—
F10	**GRAVITY ALTERATION** `GAlteration`	In **space** and in some **dungeons**, there may be no gravity. You can fly around and move in this state. Interact with the **button** in front of you to change the **gravity**.	Gravity button pressed	`Gravity` TargetPoint `GravityButton`	`StateValue`	Gravity button (`GravityButton`)
F11	**GRAVITY MOVEMENT** `GMovement`	Press **A** to take off. Use the **left stick** to move around. Push the **right stick** up/down to rise or descend, and left/right to turn.	Collider	`Plane05_5` · tag `GParkour`	`TriggerPlane`	—
F12	**AIR BRAKE** `GStop`	Press **B** to stop moving. Interact with the button to proceed to the next section.	Button pressed	TargetPoint `TrapDoor1Button`	—	Button that opens the first trapdoor (`TrapDoor1Button`)
F13	**CONTINUE UP** `TrapDoorOver`	Go to the next section.	Collider	TargetPoint `TrapDoor1Over`	—	Area above, in the zero-gravity parkour zone
F14	**ROLL** `GRoll`	Press down on the **right stick** or **left stick** to roll.	Collider	`Plane06_6` · tag `GParkour001`	`TriggerPlane`	—
F15	**CHANGE PITCH** `GPitch`	Press **X** or **Y** to change your pitch.	Collider	`Plane06_9` · tag `GParkour02`	`TriggerPlane`	—
F16	**ENABLE GRAVITY** `Stop`	Interact with the **button** to enable **gravity** and open the **trapdoor** to continue.	Button pressed	tag `stop`	—	—

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
F17	**CONTINUE THE DUNGEON** `GEnabled`	Go ahead into the dungeon. Don't worry, just jump: it won't hurt you.	Collider	`Plane07`	`TriggerPlane`	—
F18	**PERSONAL INVENTORY** `PersonalInventory`	Your **inventory** can store up to **4 items**. Look at your **left arm** and press the **inventory button** with your **right hand** to open it.	Inventory opened	—	—	Inventory button
F19	**GRAB OBJECTS & HOLOMODELS** `Grab`	To grab an object, place your hand on it and hold the **grab button**. Holding an object does not affect your **movement**. You can hold **one object per hand**, and you can still interact with other items while holding one.	Player grabs the missile launcher ammo box	tag `TutorialGrabAmmoBox` *Note: the tag is set on the ammo box actor in the dungeon interior (INT).*	`Base_GrabObjectOfClass`	Ammo boxes
F20	**PLACING LOOT IN THE INVENTORY** `PIAddLoot`	Grab an item with your **right hand**. Bring it over the **inventory button** on your **left wrist** and release it to store it automatically. Your **inventory** does not need to be open to store items.	Loot added to the inventory	—	—	Personal inventory
F21	**EXIT THE DUNGEON** `ExitDungeon`	Now that your loot is secure, it's time to bring it back to the ship. Open the dungeon's **airlock**, then step out into **outer space**.	Player leaves the dungeon	Pointer tag `TutorialShipAirlockPointer` `Airlocks_ExitDungeon`	—	Dungeon airlock
F22	**OUTER SPACE** `OuterSpace`	Move through outer space using the **zero-gravity controls**. You can travel freely in any direction, but for now, return to your **ship** to store the **loot** you just collected.	Player enters the ship (first press of the ship airlock button) or collider *Note: the trigger currently fires when the second airlock door opens, not the first; detecting the first door is harder.*	—	`PlayerEnterShip`	Ship airlock: `Ship00b` in the scene, via a scene component on its airlock (`TutorialShipAirlockPointer`)

4 Steps: ship

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
S1	**THE PATHFINDER** `Pathfinder`	Welcome aboard the **Pathfinder**, your ship for travelling among the stars. First, store your **loot** in the ship's **inventory**.	Collider	—	—	Inventory area
S2	**TAKE LOOT FROM YOUR INVENTORY** `ExtractPInventory`	Open your **personal inventory**, then use your **right hand** to grab the **ammo box**. The item is extracted automatically and placed directly into your hand.	Ammo box taken from the personal inventory	—	—	Personal inventory button
S3	**SHIP INVENTORY & AMMO** `PutInventory`	Place the ammo in the **inventory**: put it on the **inventory table** and press **Insert**. Ammo is stored for later use and may be used automatically by **weapons**. Not all **weapons** use ammo.	Ammo placed in the inventory	—	`ShipInventory`	Ship inventory table
S4	**THE COCKPIT** `Cockpit`	Now that your **loot** is stored in the **ship**, go to the **cockpit** to continue.	Player walks through the cockpit door (collider)	`Plane08` Pointer tag `cockpit`	`TriggerPlane`	Cockpit door
S5	**CAPTAIN SEAT** `Captain`	Enemies are blocking your way, so **blow them up**. Sit in the **captain seat** using the handles to access the **energy manager**.	Player sits in the captain seat	Pointer tag `captainhandle`	`StateValue`	Captain seat handles
S6	**ENERGY MANAGEMENT** `Energy`	This is the **captain's station**. Point with your **index finger** and press the **trigger** to interact with panels. Click the **squares next to a system name** to assign **energy units**. 1 yellow square = 1 unit. Energy is limited! Now assign **5 energy units** to the **weapons**.	Weapons have 5 energy units	Pointer tag `energypanel`	`Energy`	Power panel (ideally the fifth weapon power cell)
S7	**GET UP FROM THE STATION** `GetUp`	Now that the **weapons** have energy, you can start shooting. Use the **seat belt button** or press **A** on your controller to stand up.	Player leaves the captain seat	Pointer tag `captainbelt`	`StateValue`	Captain seat belt

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
S8	**GUNNER SEAT** `Gunner`	Grab the **handle** on the **gunner seat** to access the **gunner station**.	Player sits in the gunner seat	Pointer tag `gunnerhandle`	`StateValue`	Gunner seat handle
S9	**TAKE THE HEADSET** `Headset`	This is the **gunner station**. Grab the **Sol headset** and put it on to enter **turret view**.	Player enters turret view	—	`PossessPawnOfClass` (first-person turret)	Gunner headset
S10	**TURRET** `Turret`	In **turret view**, you control the **guns** to shoot enemies. Use the **left stick** to select a gun and press **A** to assign **energy**. Each gun needs a different number of **energy units**. Now assign energy to the **gatling** and/or the **missile launcher**.	Energy given to the gatling and/or missile launcher (energy pool from 5 to 0)	—	`Energy`	—
S11	**SHOOT!** `Shoot`	Now that your guns are powered, **aim** with your **head**. Use the **right stick** to rotate the **turret**. Press the **triggers** to fire the **front turrets**, and press the **sticks** to use the **side guns**. The **red bar** next to an indicator shows the enemy's health. Target indicators (red triangles and rhomboids) briefly turn **yellow** when hit. Now destroy everything in your path.	All mines destroyed	—	`AllActorsOfClassDestroyed`	—
S12	**EXIT THE TURRET VIEW** `ExitTurret`	A large enemy **ship** is warping in. Activate the **shields** and the **self-defense systems**. Grab the **headset** and take it off to exit **turret view**.	Player exits turret view	—	`PossessPawnOfClass` (spacechar)	—
S13	**EXIT THE GUNNER SEAT** `ExitGunner`	Use the **seat belt button** or press **A** on your controller to stand up.	Player leaves the gunner seat	—	—	Gunner seat belt button
S14	**AUTOMA** `Automa`	If you are alone on the **ship**, **Sol Industries** provides an **automa** for solo explorers. Activate the **automa** with the button behind the **gunner seat**. When it is active, the **guns** fire automatically at enemies in range.	Automa activated	Pointer tag `automabutton`	`StateValue`	Automa activation button
S15	**EXTRACT OBJECTS FROM THE SHIP INVENTORY** `InventoryTake`	Now activate the **self-defense**. Go to the **inventory** and open the panel. Click the **shield generator** icon to extract the shield **system chip**.	Shield generator chip extracted from the inventory	Pointer tag `inventoryscreen`	`ShipInventory` (negated)	Inventory panel (ideally the Shield button)

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
S16	**PLACE THE SHIELD SYSTEM CHIP** `ShieldChip`	In the **system panel** you can install **chips** and **mods**. Grab the **shield system chip** and insert it into its **system panel** (match the colors).	Chip placed in the system panel	Pointer tag `systempanel`	`SystemExistsInSlot ×2` *Note: there are two possible slots for the chip.*	Shield system panel
S17	**ACTIVATE THE SHIELD** `ActivateShield`	Now that the **shield** is installed, activate it. Sit in the **captain seat**.	Player sits in the captain seat	—	`Sit`	Captain seat handles
S18	**MOVE ENERGY** `MoveEnergy`	Move 3 **energy units** from the **weapons** to the **shields**. Click the second square in **weapons** to free the extra energy, then click the third square in **shields** to assign it.	Shields have 3 units, weapons keep 2	—	`StateValue`	Energy management panel
S19	**SHIELD MANAGER** `ShieldMan`	Now that the **shields** have energy, they are recharging. Grab the handle near the **holomodel** to move the shield around the **ship**. When something hits the ship, a red **triangle** appears: use it to position the **shield** and block incoming damage.	Shield handle grabbed and moved	Pointer tag `shieldminiature` *Note: the ship has only one shield emitter, so there is only one pin to move (easier to handle).*	`Base_GrabObjectOfClass`	Shield handles or the ship holomodel
S20	**THE WAY IS CLEAR, LET'S FLY** `Clear`	All enemies are defeated: it's time to leave. Move all 5 **energy units** to the **engine**, then get up from the **captain seat**.	5 energy units on the engine and player leaves the captain seat	—	`Energy`	Energy panel
S21	**PILOT SEAT** `Pilot`	Grab the **handle** on the side of the **pilot seat** to access the **pilot station**.	Player sits in the pilot seat	Pointer tag `pilothandle`	`StateValue`	Pilot seat handle
S22	**HOLOMAP MANAGER** `Holomap`	This is the **pilot station**. Use the **holomap** to explore the **universe**. Grab and move the **cursor** to zoom in and out. Click objects on the **holomap** to ping them: point with your **index finger** and hold the **trigger**. Set a course to the marked **asteroid field**.	Cursor at 50% zoom-out	`BP_Slider3d_Holomap`	`Base_GrabObjectOfClass`	Holomap cursor
S23	**SHIP GEARBOX** `Gearbox`	Grab the **sphere** to choose a movement mode: **Inertial brake:** the ship stops all movement. **Operational flight mode:** the ship moves at maximum speed. **Micro-thrust mode:** the ship moves slowly, for precise maneuvers. Set it to **operational flight mode**.	Gearbox in flight mode	—	—	Ship gearbox

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
S24	**SHIP COMMAND** `ShipMovement`	Use the **right trigger** to accelerate and the **left trigger** to decelerate. Use the **left stick** and **right stick** to rotate the **ship**. The **holomodel** shows rotation and speed. More **energy** in the **engines** increases speed. Now try flying around.	Collider	`SphereIn01` · tag `comand`	`PlayerNearTo`	Ship maneuver joysticks
S25	**REACH THE ASTEROID FIELD** `ReachAsteroids`	Reach the **asteroid field** to see if there is any **loot**. Use the **holomap** to help you find it.	Green loot indicator appears, or collider	`SphereOut01` · tag `fetch`	`PlayerNearTo`	Asteroid field
S26	**INDICATORS** `Indicators`	When you or the **ship** are near objects or enemies, an **indicator** appears. Red **triangles**: enemies, with health bars. Red **rhomboids**: enemy ship parts; destroy them for heavy damage. Green **hexagons**: **loot**. There is **loot** nearby: let's check it out.	Collider (ship or player)	`SphereOut02` · tag `indicator`	`PlayerNearTo`	Building on the asteroid
S27	**UNIVERSE EXPLORATION** `Exploration`	In a sector, you can leave the **ship** at any time to collect **loot** and explore dungeons or abandoned structures. Leave the **ship** and grab the loot on this **asteroid**.	Repair sphere grabbed	—	`Base_GrabObjectOfClass`	Repair sphere
S28	**REPAIR SPHERE** `RepairSphere`	This is a **hull repair sphere**, used to repair the **ship's hull**. Other types, like the **system repair sphere** (blue), repair **systems**. Bring the **sphere** back to the **ship**.	Player enters the ship (first press of the ship airlock button) or collider	Pointer tag `airlockbutton`	`PlayerEnterShip`	Ship airlock
S29	**REPAIR THE SHIP** `HullRepair`	With this **sphere** you can repair the **ship**. Go to the **system panel** and place it in the round hole: it will be absorbed automatically.	Ship repaired	Pointer tag `shipsyspanelrepairsphere`	`ShipHullRepaired`	Repair sphere hole in the hull panel
S30	**WARP MANAGER** `Warp`	Now that the **ship** is repaired, there is more **energy** for a warp jump. Go to the **captain seat** and assign 5 **energy units** to the **warp**, leaving some energy on the **engine**. The **warp** only works when fully powered.	5 units on the warp and at least 1 on the engine	—	`Energy`	Captain's energy panel
S31	**CHOOSE A DESTINATION** `Destination`	Leave the **captain seat** and go to the **galactic map**. Click a **sector** (a floating colored sphere) and select **Calculate coordinates**.	Coordinates calculated	Pointer tag `galaxymap`	`WarpCalcStarted`	Galactic map

#	Step	Player text	Completes when	Trigger / pointer	Task	Line target
S32	**EXIT THE ASTEROID FIELD** `ExitAsteroids`	The coordinates are set: the last step is to align the **ship** with the **warp vector**. First, leave the **asteroid field**. Avoid collisions: if anything hits the **ship** during the warp, it may explode.	Collider	`SphereIn02` · tag `getout`	`PlayerNearTo`	—
S33	**LET'S GET OUT OF HERE** `GetOut`	The path is clear: align the ship with the **luminous trail** until it turns green, then hold your position. Grab the **holo lever** and push it forward to start the **jump**.	Jump successful	—	`WarpStarted`	Warp lever
S34	**TUTORIAL COMPLETED** `Ending`	Congratulations, you have completed the tutorial. See you in space, cadet!	End of the tutorial: the player returns to the lobby	—	—	—

5 In-engine result

This is how the on-foot tutorial dungeon looks in the engine. The ship is the large structure at the end of the corridor, past the airlock.

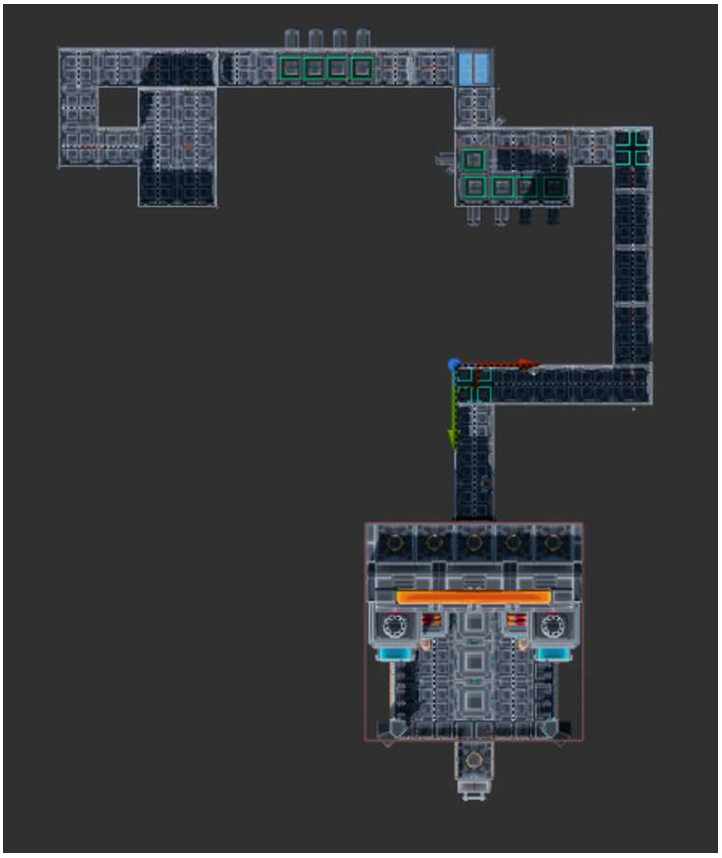


Figure 3 *Dungeon and ship, top-down view.*

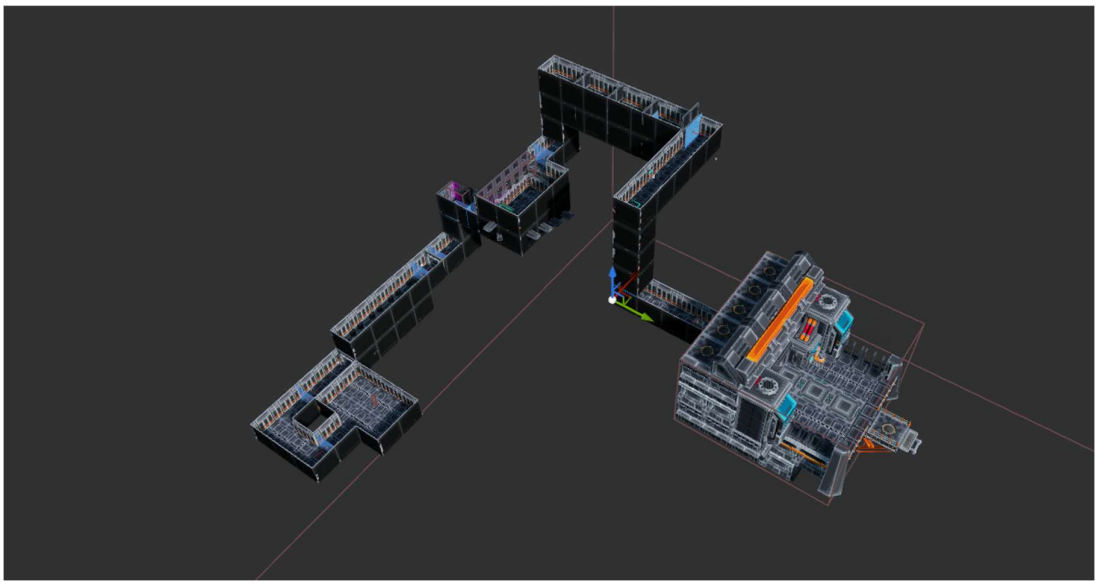


Figure 4 *Dungeon and ship, side view.*

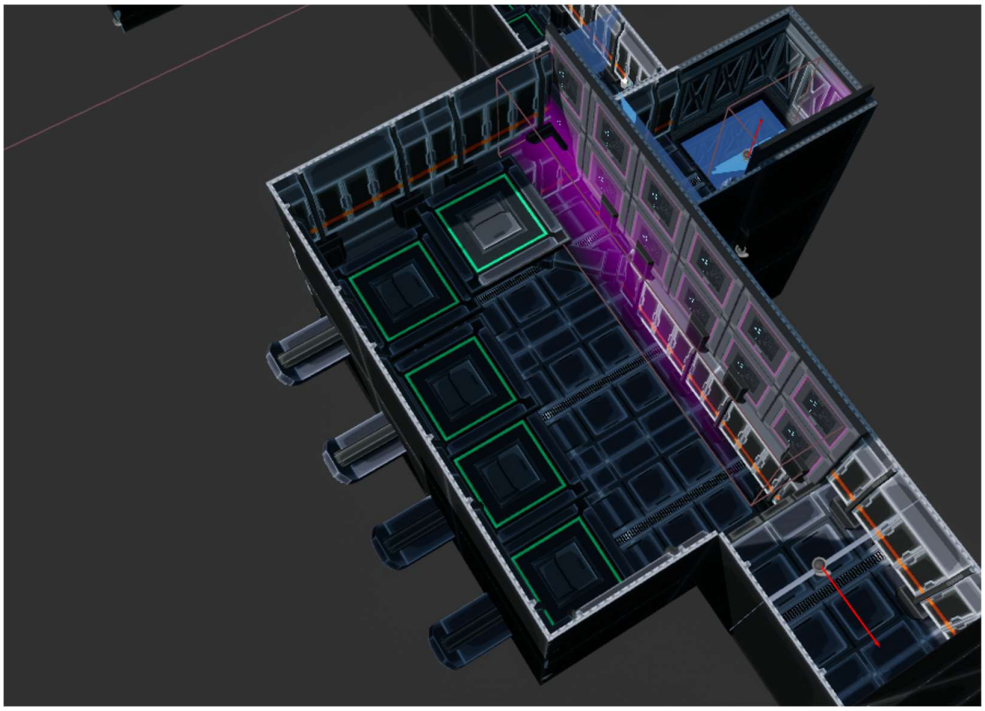


Figure 5 *Parkour room, with the jump platforms and the gravity wall.*